INCH-POUND
MS51969C
15 August 2011
SUPERSEDING
MS51969B
24 July 1972

DETAIL SPECIFICATION SHEET

NUT, PLAIN, HEXAGON – BRASS, BLACK CHEMICAL FINISH, UNC-2B

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and procurement specification FF-N-836.

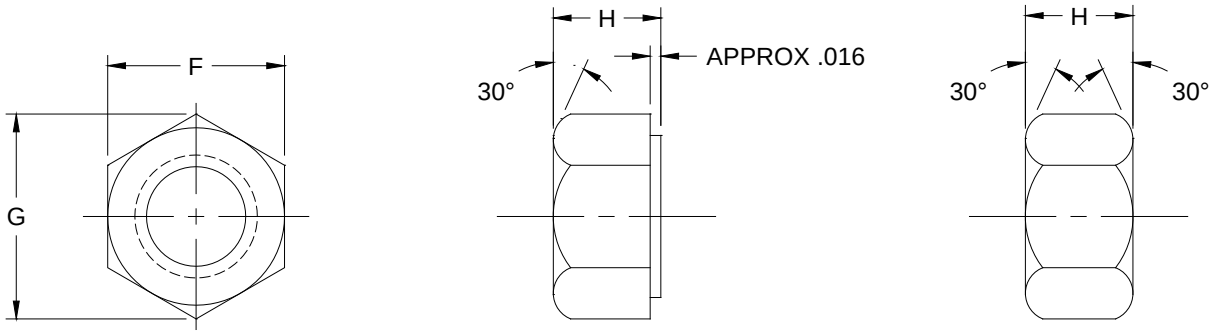


FIGURE 1. Nut, Acceptable Designs.

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TABLE I. Dash Numbers and Dimensions.

MATERIAL		DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO
BRASS		1	2	3	4	5	6	7	8	9	10
NOMINAL SIZE OR BASIC MAJOR DIAMETER OF THREAD		1/4 .2500	5/16 .3125	3/8 .3750	7/16 .4375	1/2 .5000	9/16 .5625	5/8 .6250	3/4 .7500	7/8 .8750	1 1.0000
THREADS PER INCH		20	18	16	14	13	12	11	10	9	8
F	WIDTH ACROSS FLATS NOM.	7/16	1/2	9/16	11/16	3/4	7/8	15/16	1-1/8	1-5/16	1-1/2
	WIDTH ACROSS FLATS MAX.	.4375	.5000	.5625	.6875	.7500	.8750	.9375	1.1250	1.3125	1.5000
	WIDTH ACROSS FLATS MIN.	.428	.489	.551	.675	.736	.861	.922	1.088	1.269	1.450
G	WIDTH ACROSS CORNERS MAX.	.505	.577	.650	.794	.866	1.010	1.083	1.299	1.516	1.732
	WIDTH ACROSS CORNERS MIN.	.488	.557	.628	.768	.840	.982	1.051	1.240	1.447	1.653
H	THICKNESS NOM.	7/32	17/64	21/64	3/8	7/16	31/64	35/64	41/64	3/4	55/64
	THICKNESS MAX.	.226	.273	.337	.385	.448	.496	.559	.665	.776	.887
	THICKNESS MIN.	.212	.258	.320	.365	.427	.473	.535	.617	.724	.831

MATERIAL		DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO
BRASS		11	12	13	14	15	16	17	18	19	20
NOMINAL SIZE OR BASIC MAJOR DIAMETER OF THREAD		1-1/8 1.1250	1-1/4 1.2500	1-3/8 1.3750	1-1/2 1.5000	1-3/4 1.7500	2 2.0000	2-1/4 2.2500	2-1/2 2.5000	2-3/4 2.7500	3 3.0000
THREADS PER INCH		7	7	6	6	5	4-1/2	4-1/2	4	4	4
F	WIDTH ACROSS FLATS NOM.	1-11/16	1-7/8	2-1/16	2-1/4	2-5/8	3	3-3/8	3-3/4	4-1/8	4-1/2
	WIDTH ACROSS FLATS MAX.	1.6875	1.8750	2.0625	2.2500	2.6250	3.0000	3.3750	3.7500	4.1250	4.5000
	WIDTH ACROSS FLATS MIN.	1.631	1.812	1.994	2.175	2.538	2.900	3.262	3.625	3.988	4.350
G	WIDTH ACROSS CORNERS MAX.	1.949	2.165	2.382	2.598	3.031	3.464	3.897	4.330	4.763	5.196
	WIDTH ACROSS CORNERS MIN.	1.859	2.066	2.273	2.480	2.893	3.306	3.719	4.133	4.546	4.959
H	THICKNESS NOM.	31/32	1-1/16	1-11/64	1-9/32	1-1/2	1.23/32	1-59/64	2-9/64	2-23/64	2-37/64
	THICKNESS MAX.	.999	1.094	1.206	1.317	1.540	1.763	1.970	2.193	2.415	2.638
	THICKNESS MIN.	.939	1.030	1.138	1.245	1.460	1.675	1.874	2.089	2.303	2.518

NOTES:

1. Material: Brass, Naval, in accordance with ASTM B21/B21M, Copper, Alloy UNS C46200, H04 hard or UNS C46400, H02 half-hard.

2. Protective Coating: Black chemical finish in accordance with MIL-F-495.

3. Threads: The threads shall be in accordance with FED-STD-H28/2.

4. Dimensions: All dimensions are in inches, unless otherwise specified.

5. Part Number: The MS part number consists of the MS number, plus the dash number.
   Example: MS51969-1.

6. Referenced documents shall be of the issue in effect on date of invitations for bid.

7. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

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8. Black oxide coated brass threaded fasteners are prohibited from use on Navy ships. See S9086-CJ-STM-010, Naval Ship's Technical Manual, Chapter 075, Fasteners.

9. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

The following nuts covered by dash numbers given in MS35690 are cancelled after the date indicated on the cancelled document. Use the dash numbers given in the preceding sheet. The cancelled nuts cannot always replace the new nuts and should be used until existing stocks are depleted. Use only the new nuts for design and replacement. Replacement shall be in accordance with Table II. Interchangeability information regarding part numbers on superseding drawings is also contained in Table II.

Table II. Interchangeability table of cancelled and superseding part numbers.

PART NUMBERS					
CANCELLED MS35690	NEW MS51969	CANCELLED MS35690	NEW MS51969	CANCELLED MS35690	NEW MS51969
411	1	911	6	2411	14
412	1	1011	7	2811	15
511	2	1211	8	3211	16
512	2	1411	9	3611	17
611	3	1611	10	4011	18
612	3	1811	11	4411	19
711	4	2011	12	4811	20
811	5	2211	13		

MILITARY INTEREST

Custodians:
Army – AR
Navy – SH
Air Force – 71

Preparing activity:
DLA - IS

(Project 5310-2011-011)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil

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